

Enhancing absorption performance in metal hydride hydrogen storage: Implementation of fins and a water jacket

Dae Yeob Lee, Yasser Mahmoudi, Vincenzo Spallina , Amir Keshmiri ^{*}

School of Engineering, The University of Manchester, Manchester, M13 9PL, UK

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ABSTRACT

This study investigates the flow and heat transfer characteristics of a Metal Hydride (MH) hydrogen storage device featuring internal fins and a water jacket for improved hydrogen storage performance. The MH is modelled as a porous medium, with computational fluid dynamics used to analyse transient heat and mass transfer, incorporating a conjugate turbulent heat transfer model for the water jacket. Investigations covered fin arrangements (in-line and staggered), materials (stainless-steel and copper), water flow rates ($Re = 5100$ – 22000), and hydrogen inlet pressures (5 – 20 bar). Results show that staggered copper fins and higher flow rates or pressures significantly reduce hydrogen absorption time: staggered fins with $Re = 5100$ reduce absorption time by 30% , $Re = 22000$ achieves further 27% reduction, and increasing inlet pressure to 20 bar decreased it by 61% . However, increased fin numbers and pressures introduce challenges such as pressure drop and reactor mass. Four reactor designs were assessed, highlighting trade-offs in performance.

NOMENCLATURE

Abbreviation		Greek symbols	
HTF	heat transfer fluid	β	plateau hysteresis factor
MH	metal hydride	ε	porosity
PCI	pressure-composition-isotherm	λ	thermal conductivity [W/(m K)]
Parameters		μ	viscosity [kg/(m s)]
C	reaction rate constant [1/s]	ρ	density [kg/m ³]
c_p	specific heat capacity [J/(mol K)]	τ	viscous force [Pa s]
ΔH	reaction enthalpy [J/mol]	ϕ_0	plateau flatness constant
ΔS	reaction entropy [J/(mol K)]	ϕ_s	plateau flatness factor
E	activation energy [J/mol]	Subscripts	
g	gravitational acceleration constant [m/s ²]	a	absorption
K	Permeability [m ²]	e	effective
M	molecular weight [g/mol]	emp	empty state
$\vec{n}$	unit normal vector	eq	equilibrium
P	pressure [Pa]	g	gas
R	ideal gas constant [J/(mol K)]	i	inner
Re	Reynolds number	l	liquid
r	r-coordinate [m]	m	metal
S	absorption rate [kg/(m ³ s)]	o	outer
t	time [s]	ref	reference
T	temperature [K]	sat	saturated state
u	velocity [m/s]	w	wall

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(continued)

wt.	weight percentage
%	
z	z-coordinate [m]

1. Introduction

Hydrogen offers a high energy density, serves as a clean energy source, and produces no harmful emissions [1]. In addition, given that hydrogen is the most abundant element in the universe, it is a compelling candidate for developing a new, suitable energy infrastructure [2]. However, hydrogen storage presents a significant challenge due to its extremely low density (0.089 g/L) under atmospheric conditions [3]. Compared to other energy sources, hydrogen gas exhibits a notably low volumetric energy density under these conditions, although it possesses a high energy density per mass [3,4]. Consequently, various methods, such as compression and liquefaction, are necessary to increase its volumetric energy density. These conventional techniques are straightforward to increase hydrogen density, nevertheless, they also introduce challenges related to safety and the substantial energy consumption required for these processes [5,6].

Metal hydrides (MHs) for hydrogen storage are emerging as a novel approach, offering high volumetric energy density and operation under

^{*} Corresponding author.

E-mail address: a.keshmiri@manchester.ac.uk (A. Keshmiri).

less severe conditions with enhanced safety [7]. MHs hydrogen storage utilise chemical bonds formed between hydrogen and metal or metal alloys. When hydrogen molecules reach the metal surface, they first dissociate and then hydrogen atoms diffuse into the metal lattice [8]. Since hydrogen can be stored in its atomic state, MHs hydrogen storage system can achieve a very high volumetric energy density. The mechanism of hydrogen charging and discharging is provided by absorption and desorption, respectively. The kinetics of MHs are closely related to temperature and pressure, and the direction of these gas-solid interactions is determined by the equilibrium pressure [9]. Absorption is dominant when the hydrogen gas pressure is higher than the equilibrium pressure, while desorption proceeds when the hydrogen gas pressure is lower. The absorption is an exothermic process, hence favoured at low temperature [10]. The combination of absorption and desorption makes MHs suitable for thermal applications such as hydrogen compression, thermal energy storage, and heat pumps [11–13].

Several MHs have been researched for hydrogen storage applications. The most favoured MHs are magnesium (Mg)-based hydrides, titanium (Ti)-based hydrides, and lanthanum (La)-based hydrides [9]. Mg-based hydrides offer high hydrogen storage capacity but they suffer from poor kinetics and require high temperature and pressure for operation [14]. Ti-based hydrides are attractive due to their abundance and long-term stability, however, their high equilibrium pressure, activation difficulties, and low hydrogenation rate limit their applicability [7]. Finally, La-based hydrides have a relatively low hydrogen storage capacity (1.4 wt%), however, under ambient temperature and pressure, they exhibit fast kinetics, and possess a high hydrogen volumetric capacity [15]. These favourable characteristics make La-based hydrides a promising material for various applications, and they are the most commonly utilised on an industrial scale [16]. Therefore, in this study, Lanthanum–Nickel alloy (LaNi₅) was used to investigate the absorption performance.

Overall, the storage capacity and design of appropriate equipment require substantial effort to deal with slow absorption and desorption due to the low thermal conductivity of MHs [17]. To address this issue, many studies have explored effective heat transfer methods to enhance MHs reaction rate through experiments and numerical simulations. The most fundamental approaches include installing fins to enhance heat transfer and using heat transfer fluids (HTF) in the reactor to increase the cooling and heating. MacDonald and Rowe [18] applied external fins to the reactor to reduce its thermal resistance, which effectively decreased the hydrogen desorption time. Askri et al. [19] compared the effects of internal and external fins on absorption time in an MH reactor using a mathematical model. Chung et al. [20] experimentally assessed the absorption and desorption time of an MH reactor using a central heat pipe with external fins. Both absorption and desorption times were reduced by half compared to the case without heat pipe. Chibani et al. [21] applied external fins with phase change materials in the reactor as a cooling method for large-scale hydrogen storage within an energy storage system. Bai et al. [22] optimised the absorption time and thermal resistance of MH reactor by implementing an internal longitudinal tree-shaped fin design, achieving a 21 % reduction in absorption time compared to simple radial fins. The installation of an HTF tube has generally demonstrated a substantial reduction in absorption time. Tong et al. [23] developed a reactor design incorporating various types of heat exchanger pipes, including straight and coiled-tube configurations. Although the presence of the heat exchanger reduces the quantity of stored hydrogen, it significantly decreases hydrogen absorption time by 34 % and 73 %, respectively. Parida and Muthukumar [24] developed an MH absorption bed with fins and a central cooling tube. Three fin designs, which is longitudinal, transverse, and spiral fins are compared for absorption and desorption rate. A few studies have analysed MH reactors with internal cooling fins and an external water jacket, conducting parametric studies on operating conditions, fins designs, and different cooling fluids [25,26].

Similar research was conducted by Prasad and Muthukumar [27].

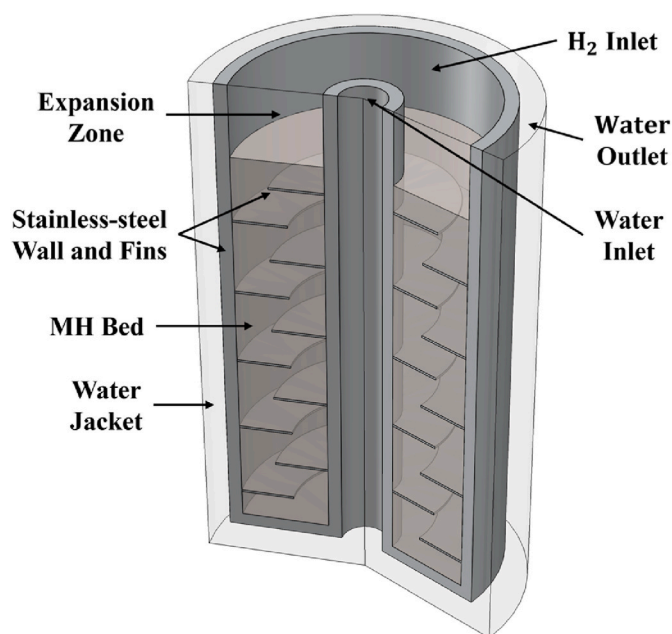


Fig. 1. Three-dimensional view of one of the MH reactor designs tested in the present study.

They compared two cases: 1) outer cooling channel, and 2) outer cooling channel with internal fins. The results of the numerical analysis showed that the second case exhibited a 51.8 % improvement in absorption time, and the peak temperature of the thermal fluid in case 1 was 35.6 °C, while in case 2, it was 41.2 °C, demonstrating the effective heat transfer achieved using internal fins.

Although several studies have considered internal and external HTF flow for effective cooling, there is still lack of some fundamental aspect of the MH modelling, as outlined below:

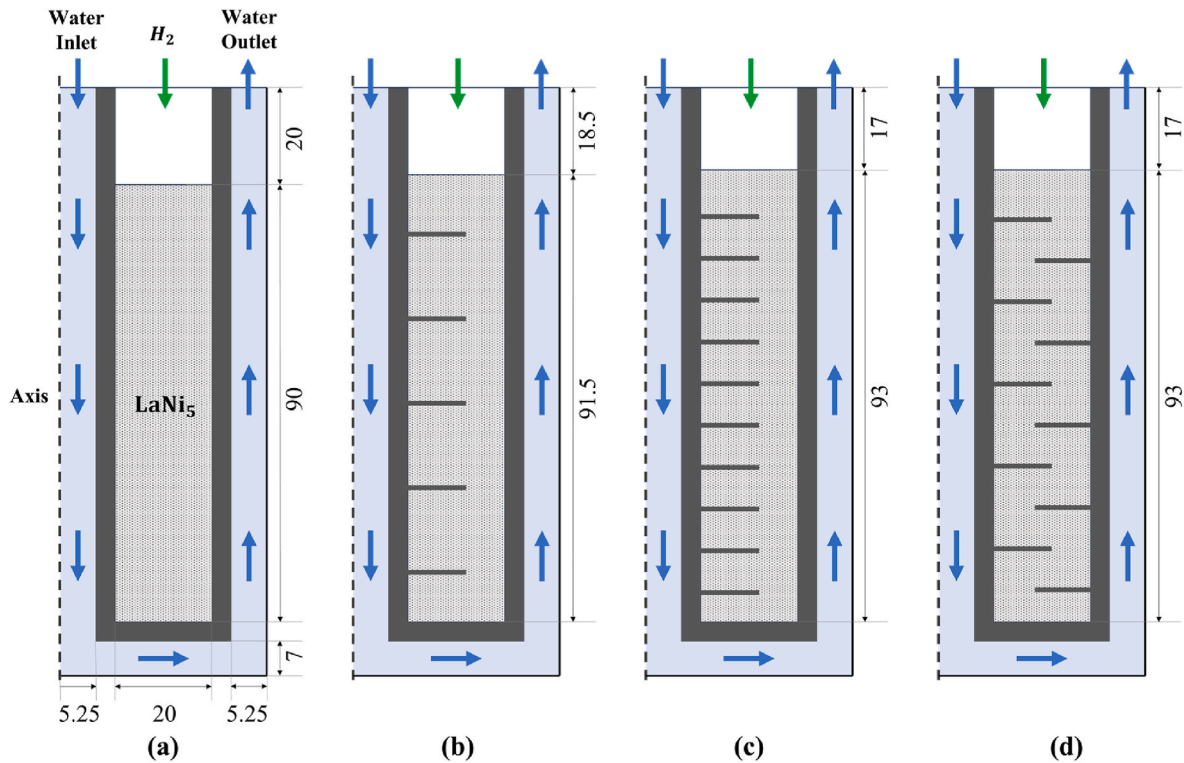
- (i) Previous studies have not incorporated a fully coupled conjugate heat transfer model for MH reaction, where heat transfer in both solid and fluid domain is simultaneously solved. This modelling approach is essential because the heat and mass transfer during absorption strongly depend on local temperature, hydrogen concentration, and pressure [28]. By direct coupling heat transfer between solid structures and surrounding fluid flow, the present study provides a more realistic and detailed representation of transient behaviour during hydrogen absorption.
- (ii) Most of the previous studies have simplified the cooling system by applying a fixed heat transfer coefficient (HTC) instead of simulating the actual cooling fluid flow. The present study conducts a transient simulation incorporating a turbulent cooling flow, allowing for a more accurate assessment of its impact on MH absorption performance. Furthermore, instead of assuming a constant HTC, HTC values are inversely derived under different conditions.
- (iii) Due to MH expansion during absorption, the simulation should include an interface between MH bed and the expansion volume—a domain of pure hydrogen gas—to secure space for expansion (Fig. 1). Moreover, incorporating the expansion zone affects the thermal flow pattern and increases thermal resistance, leading to reduced reaction rates [29].

The present study aims to address these gaps by developing a comprehensive numerical model that incorporates conjugate heat transfer and turbulence model for HTF flow in the reactor. In this paper, MH absorption for hydrogen charging was investigated using computational fluid dynamics (CFD). The established simulation model was

Table 1

Geometrical details of the reactor designs modeled in the present study.

Design No.	Expansion volume [cm ³]	MH volume [cm ³]	Fin volume [cm ³]	MH weight [g]	Surface Area by fin [cm ²]	Number of fins	Fin arrangement
Design 1	45.9	206	–	857	–	0	–
Design 2	42.4	207	2.69	861	10.9	5	In-line
Design 3	39.0	208	5.37	864	21.8	10	In-line
Design 4	39.0	206	6.88	857	28.0	10	Staggered

**Fig. 2.** Geometric details of metal hydride reactor designs (a) Design 1, (b) Design 2, (c) Design 3, and (d) Design 4. Dimensions are given in mm.

validated against experiments data, and a mesh sensitivity analysis was performed. Innovative reactor designs incorporating internal fins, and an embedded water jacket are investigated to improve the absorption performance. Specifically, four different reactor designs with varying fin numbers and fin arrangement are compared for its hydrogen storage performance. Moreover, parametric studies on water flow rates, hydrogen inlet pressures, and heat transfer fin materials were conducted to optimise the MH absorption performance under various operating conditions with the novel reactor design.

2. Problem definition

2.1. Reactor design

To investigate the cooling effect of internal fins and water jacket, four different reactor configurations are compared. The geometry of the reactor is composed of four zones: the MH bed, the expansion zone, the reactor wall with fins, and water jacket. The MH bed is a porous medium for LaNi₅ particles. The LaNi₅ alloy undergoes approximately 17 % volumetric expansion during absorption [30]. To prevent deformation of the reactor due to the stress generated in this process, an adequate amount of space is required, which is designed as the expansion zone. In the present reactor designs, 19 % of expansion volume is secured to prevent reactor deformation.

The reactor is cylindrical in shape, with water entering from the top of the reactor, flowing down to the bottom, and then rising to the top rim

of the cylinder. The length and thickness of the internal fins are 12 mm and 0.5 mm, respectively, for all designs. A geometric comparison of each design is provided in Table 1 and shown in Fig. 2. The reduction in the volume of MH due to the installation of internal fins in the reactor can affect subsequent comparisons of absorption time. To ensure consistency, all designs contain nearly the same amount of MH. The height and outer radius of the cylinder with a water jacket are 120 mm and 36.5 mm, respectively. The material of the fins and reactor wall is stainless-steel (SS316), and the wall thickness is 3 mm in all areas. The diameter of the water inlet and outlet is fixed at 10.5 mm and 5.25 mm, respectively, for all designs. A three-dimensional view of Design 4 is shown in Fig. 1 as an example. The application of fins increases the contact area between the MH and the reactor wall (e.g., 90 mm in Design 1, and 93 mm in Design 3). However, this increase in the side surface area accounts for only up to 3 % of the total surface area of the fins. Therefore, the improvement in reactor performance due to fin application is primarily influenced by the increased surface area of the fins.

2.2. Initial and boundary conditions

For the numerical model, the reactor is assumed to be in thermodynamic equilibrium at the initial state:

$$T_{\text{initial}} = T_{\text{MH},0} = T_{\text{water},0} = 293 \text{ K}, P_{\text{H}_2,\text{initial}} = 1 \text{ bar} \quad (1)$$

The initial temperature of both the reactor and the water jacket is uniformly set to 293 K. For the absorption process, the initial pressure

Table 2

Initial condition of simulation.

Description	Symbol	Value	Unit
Initial temperature of MH bed	T_{initial}	293	K
Initial pressure in reactor	$P_{H_2, \text{initial}}$	1	bar
Hydrogen inlet temperature	$T_{H_2, \text{inlet}}$	293	K
Water initial temperature	$T_{\text{water, initial}}$	293	K
Water inlet temperature	$T_{\text{water, inlet}}$	293	K
Initial metal density	$\rho_{\text{m, initial}}$	8300.5	kg/m ³
Porosity of metal powder	ε	0.5	–

Table 3

Operating conditions for the parametric studies.

Description	Baseline	Values	Unit
Number of fins	0	0, 5, 10	–
Reynolds number of water	5100	5100, 7600, 14000, 22000	–
Hydrogen inlet pressure	10	5, 10, 15, 20	bar

was set at 1 bar, as the hydrogen gas pressure must be higher than the equilibrium pressure. During the simulation, the hydrogen and water inlet temperature are maintained at 293 K. The hydrogen inlet pressure is linearly increased over the first 100 s until it reaches setting pressure, after which it is maintained at this pressure until the end of the simulation. The overall initial conditions for the simulation are summarised in Table 2.

For the parametric study of water mass flow rate, values of 0.03, 0.05, 0.10, and 0.15 kg/s are selected, which correspond to the Reynolds number (Re) of 5100, 7600, 14000, and 22000, respectively. The Re is calculated as $Re = \frac{\rho u L}{\mu}$ where ρ is the density of liquid water, u is the velocity of water, L is the diameter of water jacket channel, and μ is the viscosity of liquid water. To prevent reversed flow at the water outlet due to gravity, the water flow rate is maintained above Re of 5100. The parameters for the operating conditions used in the simulation are presented in Table 3.

For the base and lateral surface of the heat transfer fluid channel outer wall, adiabatic conditions are applied:

$$-\lambda_{\text{water}} \frac{\partial T_{\text{water}}}{\partial \vec{n}} = 0 \quad (2)$$

where λ_{water} is the thermal conductivity of water, T_{water} is the temperature of the water, and $\vec{n}$ is the unit normal vector. This implies no heat transfer occurs across the base and lateral surface of the water jacket outer wall.

At the interface between MH and the reactor wall, the heat flux is expressed as:

$$-\lambda_e \frac{\partial T_{\text{MH}}}{\partial \vec{n}} = -\lambda_{\text{steel}} \frac{\partial T_{\text{w, i}}}{\partial \vec{n}} \quad (3)$$

where λ_e and λ_{steel} are the effective thermal conductivity in MH bed and thermal conductivity of stainless-steel, and T_{MH} and $T_{\text{w, i}}$ is the temperature of MH bed and inner wall.

The convective boundary condition is applied from water flow on the reactor wall:

$$-\lambda_{\text{steel}} \frac{\partial T_{\text{w, o}}}{\partial \vec{n}} = h(T_{\text{w, o}} - T_{\text{water}}) \quad (4)$$

where $T_{\text{w, o}}$ is the temperature of outer wall, and h is the heat transfer coefficient.

Table 4

Thermal and physical properties of materials [31–33].

Description	Symbol	Value	Unit
Hydrogen free metal density	$\rho_{\text{m, emp}}$	8300	kg/m ³
Saturated metal density	$\rho_{\text{m, sat}}$	8417.5	kg/m ³
Stainless-steel density	ρ_{steel}	8000	kg/m ³
Liquid water density	ρ_{water}	998.2	kg/m ³
Copper density	ρ_{copper}	8978	kg/m ³
Molecular mass of hydrogen	M_{H_2}	2.016	kg/kmol
Activation energy for absorption	E_a	21179.6	J/mol
Reaction rate constant	C_a	59.187	1/s
Reaction enthalpy	ΔH	–30800	J/mol
Reaction entropy	ΔS	–108	J/(mol K)
Specific heat capacity of metal hydride	C_{pm}	419	J/(mol K)
Specific heat capacity of hydrogen gas	C_{pg}	14890	J/(mol K)
Specific heat capacity of stainless-steel	C_{steel}	490	J/(mol K)
Specific heat capacity of liquid water	C_{water}	4182	J/(mol K)
Specific heat capacity of copper	C_{copper}	381	J/(mol K)
Thermal conductivity of metal	λ_{m}	3.18	W/(m K)
Thermal conductivity of hydrogen	λ_{g}	0.1672	W/(m K)
Thermal conductivity of stainless-steel	λ_{steel}	16.3	W/(m K)
Thermal conductivity of liquid water	λ_{water}	0.6	W/(m K)
Thermal conductivity of copper	λ_{copper}	387.6	W/(m K)
Permeability of metal powder	K	10^{-8}	m ²
Viscosity of hydrogen	μ_{g}	8.411×10^{-6}	kg/(m s)
Viscosity of liquid water	μ_{l}	1.003×10^{-3}	kg/(m s)
Plateau flatness factor	ϕ_s, ϕ_0	0.038, 0	–
Plateau hysteresis factor	β	0.137	–
Reference pressure	P_{ref}	1	bar

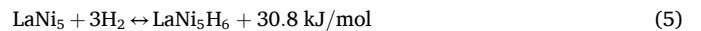
3. Numerical modelling

3.1. Assumptions

The geometry of the reactor is composed of 4 zones including: 1) the MH bed, 2) the expansion zone, 3) the reactor wall with fins, and 4) the water jacket. A two-dimensional axisymmetric model is established for numerical modelling of the cylindrical reactor. Due to the complexity of the chemical reactions during MH absorption, the following assumptions are applied in the present simulations, following the works reported in:

- (1) Local thermal equilibrium in the porous zone is applied [30,33, 34].
- (2) Hydrogen is regarded as an ideal gas [30,33,34].
- (3) Thermal-physical properties remain constant during the reaction [33].
- (4) Volumetric change of the MH during absorption is negligible [35].
- (5) Radiative heat flux is neglected [33,36].

The thermal and physical material properties of LaNi₅, hydrogen, reactor, and water are provided in Table 4. The absorption of LaNi₅ is determined by the following reaction:



3.2. Governing equations

In this section, the governing equations for the hydrogen flow in the MH bed, water flow in the jacket, chemical reaction of the MH absorption, and heat transfer for exothermic reaction and cooling are provided.

3.2.1. Momentum equations

The momentum equation for isotropic porous medium (MH bed) includes a viscous loss term. As the flow in porous medium is laminar, the Darcy's law is used. The gravity term is applied in the axial direction.

For radial momentum:

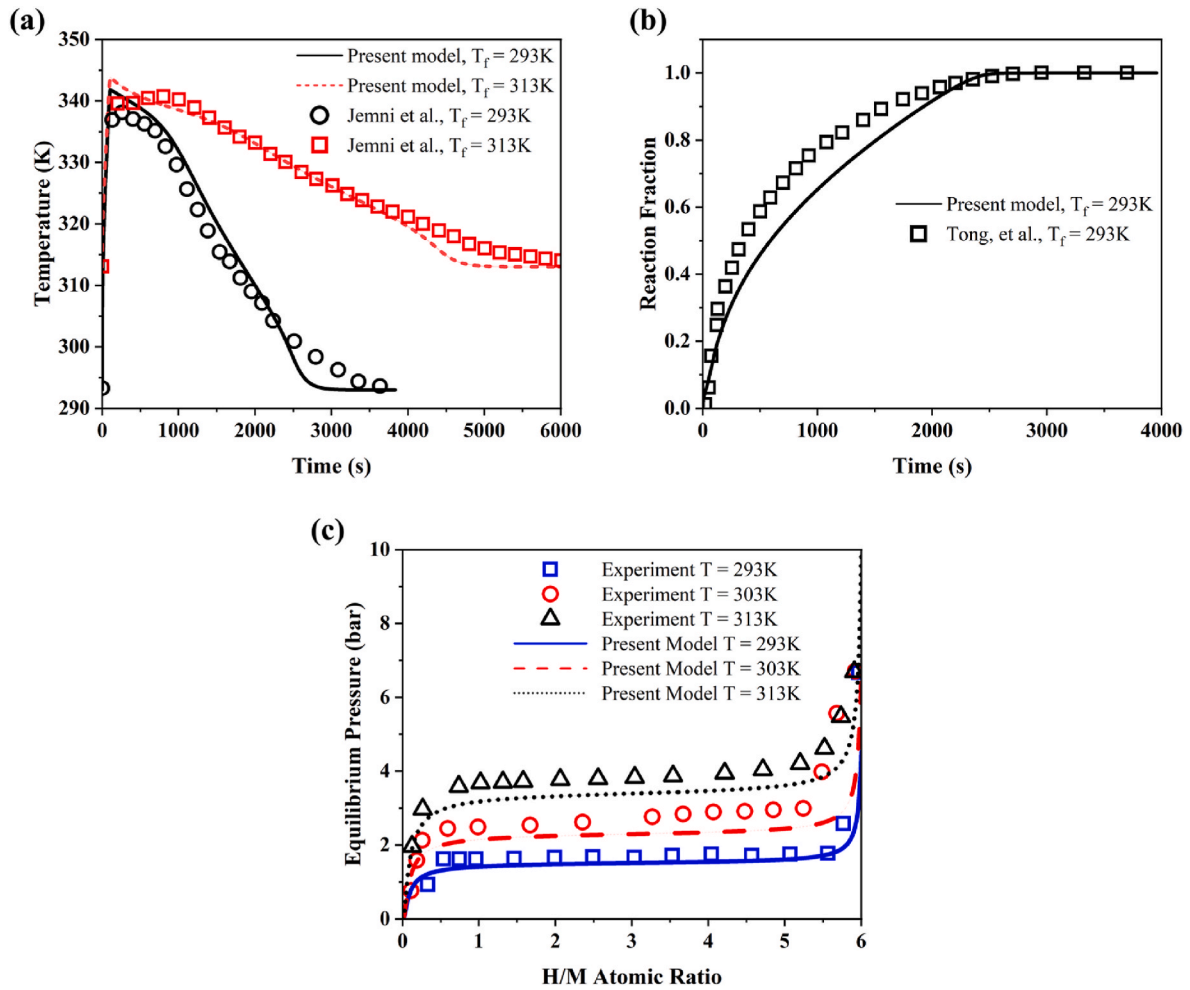


Fig. 3. Validation of numerical model: (a) Temperature profile [39], (b) Reaction fraction profile [23], and (c) PCI curves of LaNi₅ absorption reaction [39].

$$\frac{1}{\varepsilon} \left[\frac{\partial(\rho_g u_r)}{\partial t} + \frac{1}{r} \left(\frac{\partial(r \rho_g u_r u_r)}{\partial r} + \frac{\partial(\rho_g u_r u_z)}{\partial z} \right) \right] = \frac{\partial P}{\partial r} + (\nabla \cdot \tau)_r + S_{u_r} \quad (6)$$

For axial momentum:

$$\frac{1}{\varepsilon} \left[\frac{\partial(\rho_g u_z)}{\partial t} + \frac{1}{r} \left(\frac{\partial(r \rho_g u_z u_r)}{\partial r} + \frac{\partial(\rho_g u_z u_z)}{\partial z} \right) \right] = \frac{\partial P}{\partial z} + (\nabla \cdot \tau)_z + \rho_g g_z + S_{u_z} \quad (7)$$

where, ε is the porosity of MH, u is the velocity of hydrogen gas, τ is the viscous force, g_z is the gravitational acceleration, S_u is the momentum source term for porous medium, which is determined by Darcy's law [37]:

$$S_{u_r} = -\frac{\mu_g}{K} u_r, S_{u_z} = -\frac{\mu_g}{K} u_z \quad (8)$$

where μ_g is the dynamic viscosity of hydrogen, and K is the permeability. A no-slip velocity boundary condition is applied to the reactor wall and water jacket.

For the water flow in the water jacket, the Re varies from 5100 to 22000 depending on the mass flow rate, indicating turbulent flow. Given the importance of accurately analysing heat transfer between the reactor wall and the water flow, the shear stress transport (SST) $k-\omega$ turbulence model is selected due to its capability in capturing near-wall flow dynamics and thermal predictions [38].

3.2.2. Continuity equations for the MH absorption

During absorption, mass transfer occurs between the MH and

hydrogen gas in the porous media. In the MH bed, the mass of MH increases while hydrogen gas decreases during the absorption.

For the MH:

$$(1 - \varepsilon) \frac{\partial \rho_m}{\partial t} = S_m \quad (9)$$

For Hydrogen:

$$\varepsilon \frac{\partial \rho_g}{\partial t} + \frac{1}{r} \frac{\partial(r \rho_g u_r)}{\partial r} + \frac{\partial(\rho_g u_z)}{\partial z} = -S_m \quad (10)$$

where, ε is the porosity of MH, ρ_m and ρ_g denotes MH and hydrogen gas density, respectively. u is the velocity of gas. The hydrogen density (ρ_g) follows the ideal gas law:

$$\rho_g = \frac{P_g M_{H_2}}{RT} \quad (11)$$

where, P_g is the hydrogen gas pressure, and M_{H_2} is the hydrogen molecular weight which is 2.016 g/mol.

Here, S_m represents the hydrogen absorption rate per unit volume [kg/m³s], which is determined by the reaction kinetics of LaNi₅ as follows [39]:

$$S_m = C_a \exp\left(-\frac{E_a}{RT}\right) \ln\left(\frac{P_g}{P_{eq}}\right) (\rho_{m,sat} - \rho_m) \quad (12)$$

where, C_a is the reaction rate constant for LaNi₅ absorption, E_a is the activation energy, T is the MH bed temperature, and $\rho_{m,sat}$ denotes

saturated MH density after absorption. P_{eq} is the equilibrium pressure of LaNi_5 absorption.

3.2.3. Equilibrium pressure

The equation for the equilibrium pressure of LaNi_5 absorption follows the Van't Hoff equation, with an additional empirical term to account for the complex behaviour during absorption. Based on experimental results of the pressure-concentration-isotherm (PCI), the plateau flatness factor, hysteresis factor, and hydrogen concentration terms are considered to fit the experimental results (Eq. (13)) [40]. The PCI curve for the equilibrium pressure of the experiment [39] and the equation are shown in Fig. 3 (c).

$$\ln\left(\frac{P_{eq}}{P_{ref}}\right) = \left(\frac{\Delta H}{RT} - \frac{\Delta S}{R}\right) + (\phi_s + \phi_0) \tan\left[\pi\left(\frac{c}{c_{sat}} - \frac{1}{2}\right)\right] + \frac{\beta}{2} \quad (13)$$

where, P_{ref} is the reference pressure, ΔS is the reaction entropy, ΔH is the reaction enthalpy, R is the ideal gas constant, T is the temperature of MH bed, ϕ_s and ϕ_0 are plateau flatness factors, β is the plateau hysteresis factor, and c represents the concentration of MH:

$$c = \frac{2(\rho_m - \rho_{m,emp}) / M_{H_2}}{\rho_{m,emp} / M_{LaNi_5}} \quad (14)$$

where, ρ_m and $\rho_{m,emp}$ denotes MH density and hydrogen free MH density, respectively. M_{LaNi_5} represents molecular weight of LaNi_5 which is 432 g/mol.

3.2.4. Energy equations

The energy conservation in the MH bed is:

$$(\rho c_p)_e \frac{\partial T}{\partial t} + \rho_g c_{pg} \left(u_r \frac{\partial T}{\partial r} + u_z \frac{\partial T}{\partial z} \right) = \frac{1}{r} \frac{\partial}{\partial r} \left(r \lambda_e \frac{\partial T}{\partial r} \right) + \frac{\partial}{\partial z} \left(\lambda_e \frac{\partial T}{\partial z} \right) + S_m \left[\frac{\Delta H}{M_{H_2}} + T(c_{pg} - c_{pm}) \right] \quad (15)$$

where, c_p is the specific heat capacity, λ_e is the effective thermal conductivity, and S_m is the hydrogen absorption rate.

For the energy conservation of the MH reaction, the thermal equilibrium condition between MH and hydrogen gas is applied [41]. The effective thermal heat capacity and effective thermal conductivity are determined as:

$$(\rho c_p)_e = \varepsilon \rho_g c_{pg} + (1 - \varepsilon) \rho_m c_{pm} \quad (16)$$

And

$$\lambda_e = \varepsilon \lambda_g + (1 - \varepsilon) \lambda_m \quad (17)$$

where, λ_g and λ_m are the thermal conductivity of hydrogen gas and MH, respectively.

Conjugate heat transfer is solved for the fins and the reactor wall using Eq. (18)

$$(\rho c_p)_{steel} \frac{\partial T}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left(r \lambda_{steel} \frac{\partial T}{\partial r} \right) + \frac{\partial}{\partial z} \left(\lambda_{steel} \frac{\partial T}{\partial z} \right) \quad (18)$$

where, λ_{steel} is the thermal conductivity of steel wall and fins.

3.3. Numerical settings

In this study, a numerical simulation of MH absorption is conducted using the commercial CFD software ANSYS Fluent 2022R1. The governing equations from Eq. (12)–(15) are employed to analyse transient mass and heat transfer during MH absorption. The mass and energy source terms are implemented using User-Defined-Functions (UDF) in the solver. Due to the cylindrical shape of the reactor, a two-dimensional axisymmetric model is applied for the simulation. The time step is set to

Table 5

Mesh independence assessment.

Mesh	No. of cell	Average MH bed temperature [K]	90 % Absorption time [s]
Coarse	120,000	303.2	690
Medium	210,000	304.4	699
Fine	300,000	305.0	701
Extra Fine	430,000	305.4	702

10 ms for all simulation, and the residuals for all variables are converged below 10^{-6} .

For accurate simulation, a structured quadrilateral mesh is used, ensuring that the wall-adjacent cells achieved a y^+ value of 1 for the highest Re . A mesh independency is tested with four different cell numbers: 120,000, 210,000, 300,000, and 430,000. The average MH bed temperature and 90 % of the absorption time is compared for the four different meshes (Table 5). As a result, the difference in absorption time between the fine and extra fine meshes is only 0.3 %. Therefore, the fine mesh is selected for further simulations as it provides good accuracy while being computationally efficient. One simulation case took around 40 hours using 16 cores CPU (Intel E5-2680).

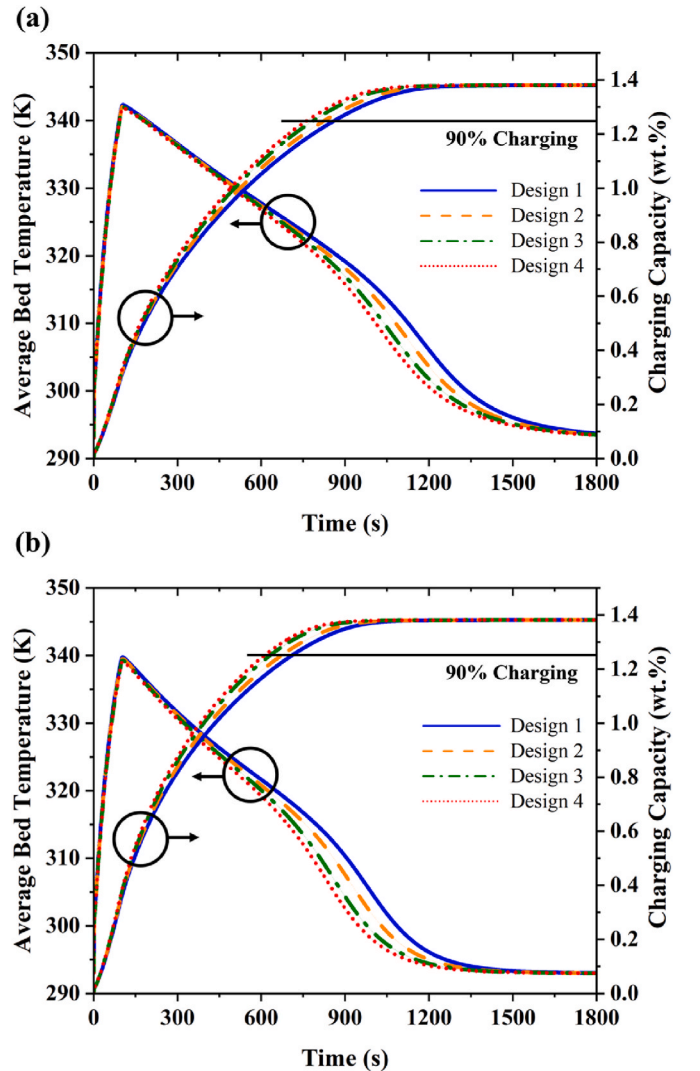


Fig. 4. Temperature and hydrogen charging capacity profiles during absorption for different reactor designs: (a) $Re = 5100$, and (b) $Re = 22000$.

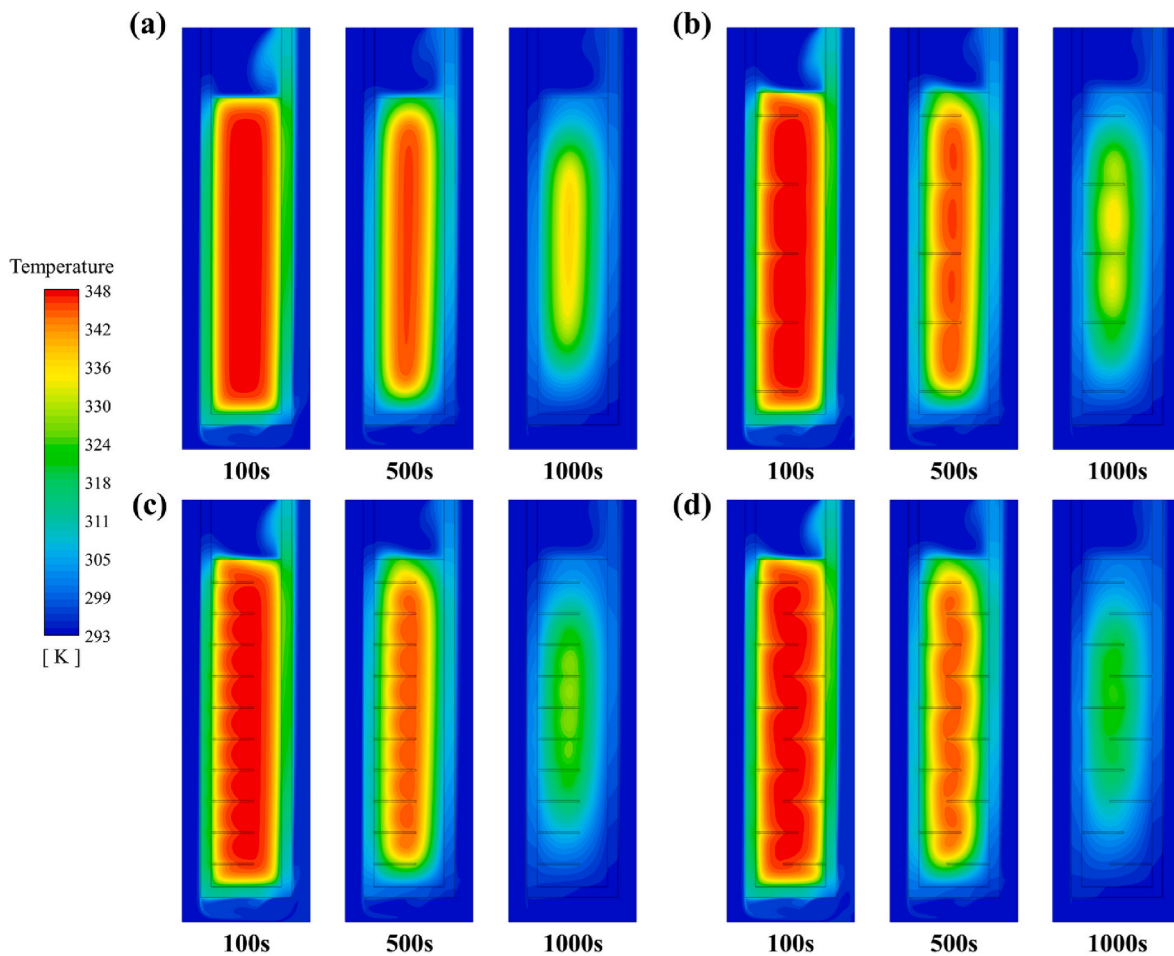


Fig. 5. Temperature contour of the four different geometry designs at 100, 500, and 1000 s: (a) Design 1, (b) Design 2, (c) Design 3, and (d) Design 4.

4. Results and discussion

The results of the numerical simulations are presented in this section. The equilibrium pressure is validated against experimental results. Additionally, temperature and reaction fraction profiles during the absorption are compared with experimental data. Hydrogen storage capacity is expressed in terms of the weight percentage (wt.%) of stored hydrogen relative to MH:

$$\text{wt.\%} = \frac{\text{Mass of stored hydrogen}}{\text{Mass of metal alloy}} \times 100 \% \quad (19)$$

4.1. Validation

The equilibrium pressure equation for LaNi_5 absorption (Eq. (13)) is compared with experimental data. Three PCI curves under different isothermal temperature conditions are shown in Fig. 3 (c). The equation for equilibrium pressure and experimental results shows good agreement under the three different conditions.

The present numerical model is validated against experimental data and a numerical model from Jemni et al. [39]. In the experiment, temperatures were measured using thermocouples during the absorption reaction at two different temperature conditions (293 K and 313 K). These experimental temperature results are compared with the predictions of the present numerical model, as shown in Fig. 3 (a). Due to limited information in the experimental literature, the reaction fraction profile is compared with another numerical model that investigated the identical experiment using numerical simulation (Fig. 3 (b)) [23].

The maximum error for temperature and reaction fraction in Fig. 3 is

2 % and 6 %, respectively. These results demonstrate that both temperature and reaction fraction profiles from the present numerical model agree well with experimental data and the literature, validating the model's accuracy.

4.2. Reactor design

In this section, four different reactor designs are tested to evaluate the effects of cooling and charging capacity variation. Water flow at $Re = 5100$ and 22000 , were tested for comparison.

The mass-averaged temperature and charging capacity variation profiles of MH bed for the different reactor designs are shown in Fig. 4. At the beginning of the absorption process, the temperature of the MH bed increases rapidly due to the exothermic hydrogen absorption reaction. The average bed temperature peaks at 100 s, which corresponds to the end of the hydrogen inlet pressure increase. The maximum average temperature is similar across all designs. Subsequently, the temperature gradually decreases due to the cooling effect of the water flow. After reaching the maximum charging capacity (1.38 wt%), the MH bed temperature decreases to 293 K, which is equal to the hydrogen and water inlet temperatures.

The impact of the cooling, influenced by reactor geometry, is assessed by comparing the time required to charge 90 % of the hydrogen in the MH. Increasing surface area using internal fins reduces thermal resistance within the reactor, enhancing heat dissipation during the exothermic absorption reaction. As a result, hydrogen charging time is shortened. Design 3 reduced the hydrogen charging time by 11 % and 6 % compared to Designs 1 and 2, respectively. Reactor Design 4, which has a staggered fin arrangement, achieved a 3.2 % reduction in

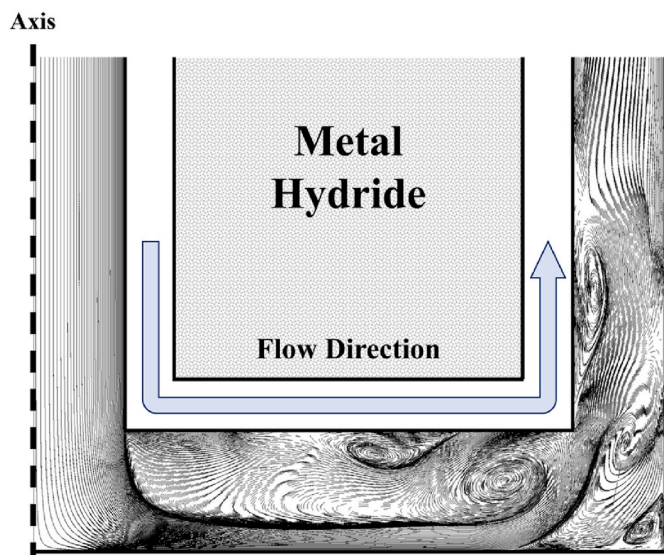


Fig. 6. Water flow streamline in water jacket ($Re = 5100$).

hydrogen charging time compared to Design 3, despite having the same number of fins. The improvement is attributed to the staggered configuration, which enhances the surfaces area.

The temperature contours at 100, 500, and 1000 s of absorption are shown in Fig. 5. At 100 s, the majority of the MH bed area reaches the

highest temperature of 348 K across all reactor designs. In all cases, the temperature starts to decrease near the reactor wall and fins. The peak temperature also decreases over time as the reaction progresses. It is noteworthy that the fins contribute to cooling the inside of the reactor, leading to lower temperatures near the fins. At 1000 s, the highest temperatures in the centre of the MH bed are 339 K, 338 K, 333 K, and 330 K for Designs 1 to 4, respectively.

At the 90° elbow of the central water jacket, a high water temperature is observed in all cases at 100 s (Fig. 5), corresponding to the mainstream water flow. After 500 s, while the temperature in the central part of the MH bed remains high, the temperature around the wall and fins decreases. In addition, the temperature variation of the water is not clearly visible, as the stainless-steel wall is cooled down by the water flow.

Moreover, due to the presence of two 90° elbows in the water jacket, a recirculation zone forms on the lower side of the cylinder (Fig. 6). In this recirculation zone, flow separation and reversed flow occur, enhancing heat transfer from the heated reactor wall by promoting fluid mixing. As a result, the temperature near the recirculation zone demonstrates lower than the lower central part of the cylinder.

The charging capacity contours, and hydrogen streamlines of the four different reactor designs are shown in Fig. 7. As the temperature decreases near the wall and internal fins (as shown in Fig. 5), the charging capacity increases in these regions. The tendency of hydrogen charging is strongly influenced by proximity to the wall and internal fins.

At 100 s, hydrogen charging capacity increases near the wall and fin areas due to its relatively low temperature, which enhances the driving

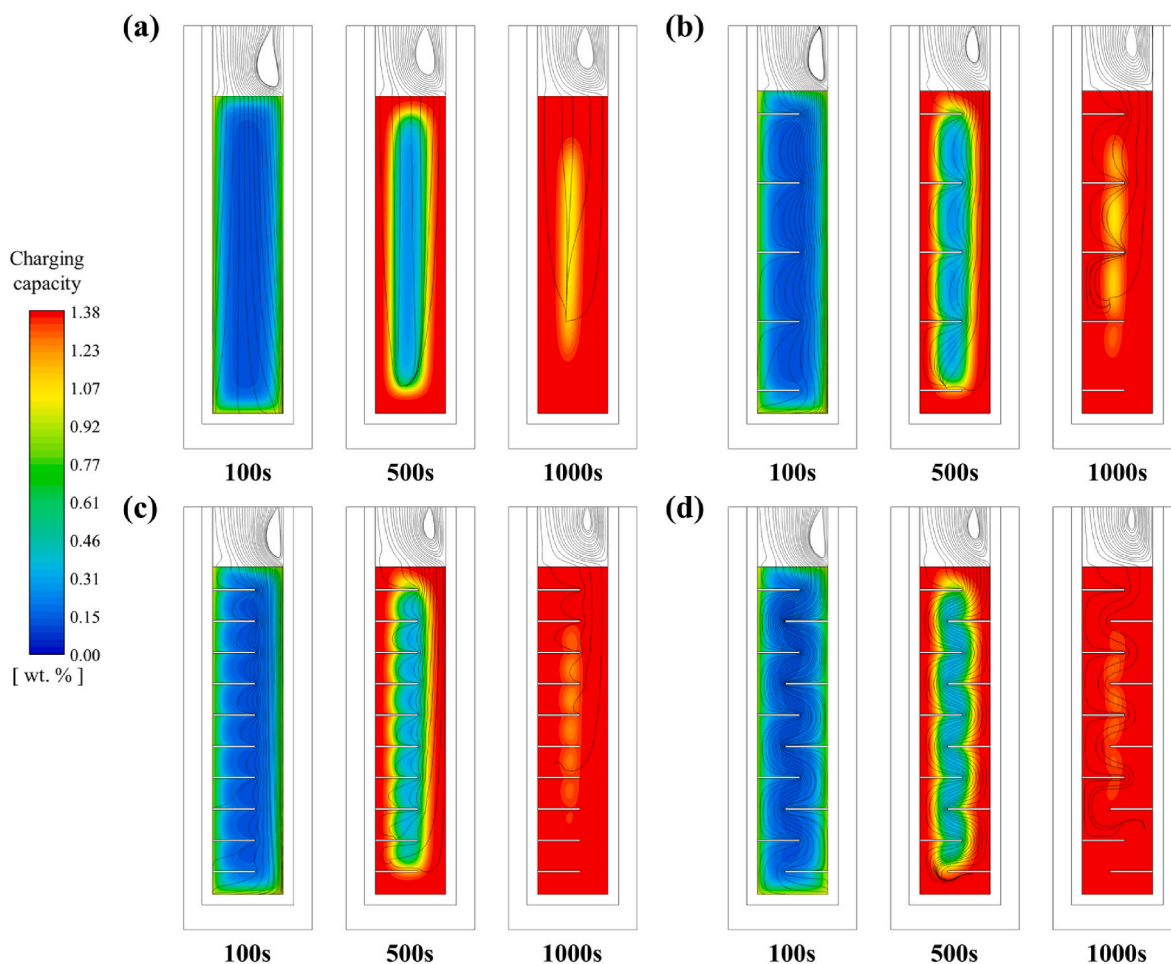


Fig. 7. Charging capacity contours and hydrogen pathlines for the four different geometry designs at 100, 500, and 1000 s: (a) Design 1, (b) Design 2, (c) Design 3, and (d) Design 4.

Table 6

Charging time [s] by different designs and water flow rates.

Reynolds Number	Design 1	Design 2	Design 3	Design 4
5100	851	816	777	754
7600	793	757	717	696
14000	728	691	651	631
22000	699	663	622	602

force for absorption. Consequently, hydrogen pathlines in the MH bed converge towards these regions, increasing charging capacity. In the expansion zone, hydrogen gas primarily flows downward into the MH bed. However, along the side walls of the reactor, where the temperature is higher, the hydrogen gas moves upward due to thermal buoyancy. At 500 s, the fully charged area expands towards the centre of the MH bed. Designs 2, 3, and 4, highlight the influence of cooling fins on hydrogen charging. At this stage, hydrogen gas no longer flows along the wall but instead moves towards the interface between the high- and low-charging capacity regions, particularly near the central of the reactor. By 1000 s, incompletely charged areas remain at the centre of the MH bed, showing differences in charging rates across the reactor designs, especially cooling fins, on hydrogen absorption.

Notably, the bottom side of the MH bed shows faster absorption than the central part of the MH bed, as the recirculation zone promotes heat transfer in this region. Consequently, this increased heat transfer lowers the equilibrium pressure, and the significant difference between the hydrogen gas pressure and the equilibrium pressure drives faster absorption.

The results of charging time for all cases are compared in Table 6. Reactor Design 4 shows the shortest hydrogen charging time in all water flow rate cases. This improvement suggests that the staggered fin arrangement enhanced heat transfer efficiency more effectively than the in-line fin configuration.

4.3. Heat transfer fluid mass flow rate

In this section, the sensitivity analysis of different reactor designs to varying Reynolds number (Re) of water is investigated. Fig. 8 (a) shows the profiles of average MH bed temperature, water outlet temperature, and charging capacity. Initially, these profiles resemble those from the reactor design study. However, variations in the Re of the water flow introduce some differences. A higher Re reduces the peak temperatures in both the MH bed and the water outlet, demonstrating that increased HTF velocity enhances heat removal more effectively than adjusting the number of heat transfer fins. Specifically, the highest MH bed temperatures for Re of 5100 and 22000 are 342 K and 340 K, respectively, while

the highest water outlet temperatures are 299 K and 295 K, respectively. This reduction occurs because higher Re leads to shorter contact time between water and the MH bed, limiting heat absorption per unit of flow and keeping the water temperature lower.

The time required to achieve 90 % hydrogen charging is analysed for different Re . As Re increased from 5100 to 7600, 14000, and 22000, the charging time decreased from 851 to 793, 728, and 699 s, corresponding to reduction of 7 %, 14 %, and 18 %, respectively.

The average heat transfer coefficient (HTC) between the stainless-steel wall and water jacket is shown in Fig. 8 (b). As Re increases, HTC rise by 20.7 %, 34.4 %, and 18.3 % at each stage. Fig. 8 (b) establishes a quantitative relationship between Re and average HTC, aiding reactor performance analysis. Notably, the largest increase in HTC occurs when Re increases from 7600 to 14000.

The charging capacity distribution along the axial middle line of the reactor is shown in Fig. 9. In Fig. 9, the x-axis represents the distance from the interface between the MH bed and expansion zone to the bottom of the MH bed (line A-A). Across all designs, the hydrogen charging capacity is higher at the interface and the bottom of the MH bed compared to its centre. In Design 1, which has no fins, the charging capacity profile is smooth. In contrast, Designs 2, 3, and 4 show higher charging capacities at the fin locations due to the presence of fins, which result in lower temperatures near this area. The reduced temperature lowers the equilibrium pressure, consequently, the greater difference between the equilibrium pressure and the loading pressure enhances the driving force (reaction rate) for absorption.

Overall, the charging capacity at the bottom of the reactor was higher than at the top, which can be attributed to the more effective heat transfer occurring at the bottom of the water jacket by the presence of recirculation of flow. The higher efficiency in heat removal at the bottom enhances the reaction rate, leading to increased hydrogen absorption in that region.

4.4. Hydrogen inlet pressure

In this section, the impact of different hydrogen inlet pressures on the hydrogen charging time and MH bed temperature are examined. The simulation is conducted using reactor Design 4 and $Re = 22000$, which showed the shortest hydrogen charging time in previous results. For the parametric study, hydrogen inlet pressures of 5, 10, 15, and 20 bar are tested. The initial hydrogen inlet pressure is set at 1 bar and linearly increased up to the target pressure within the first 100 s of the simulation. Fig. 10 shows the average MH bed temperature and charging capacity profiles for the different hydrogen inlet pressures.

The effect of hydrogen inlet pressure shows a different trend

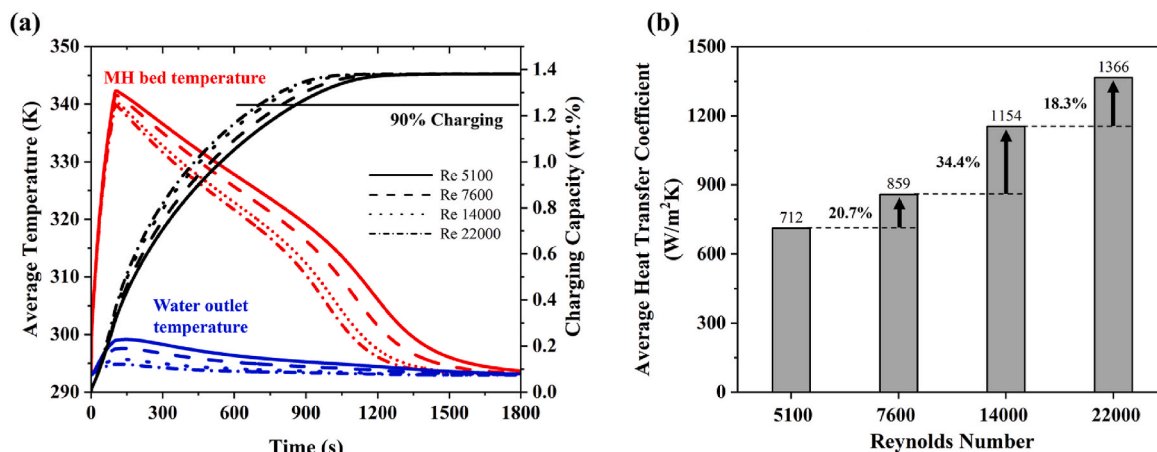


Fig. 8. (a) Temperature and hydrogen charging capacity profiles during absorption for different Reynolds number of Design 1, and (b) Average HTC between SS wall and water jacket.

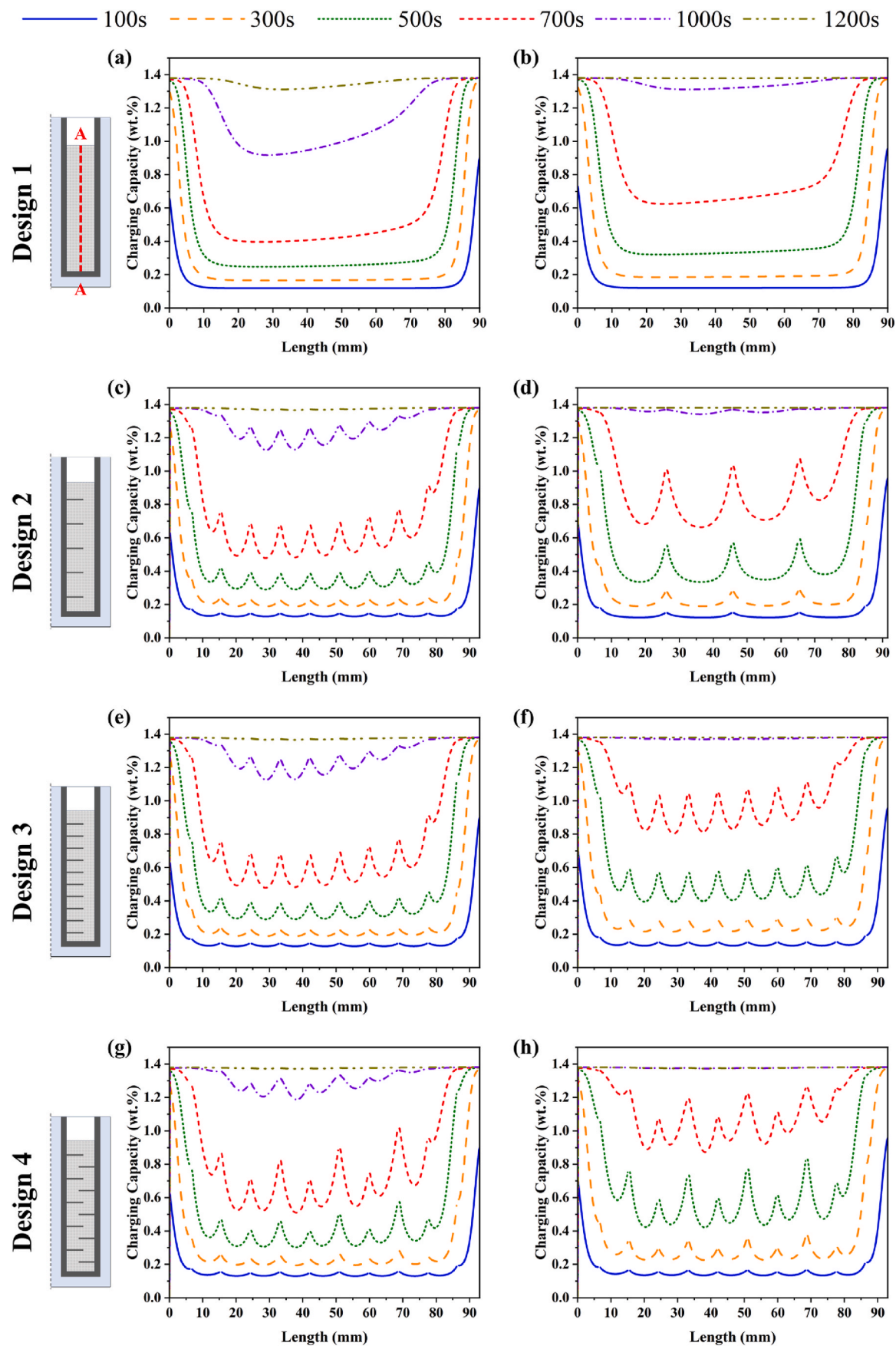


Fig. 9. Charging capacity curves at the middle line of the reactor (line A-A): (a, c, e, g) $Re = 5100$, (b, d, f, h) $Re = 22000$.

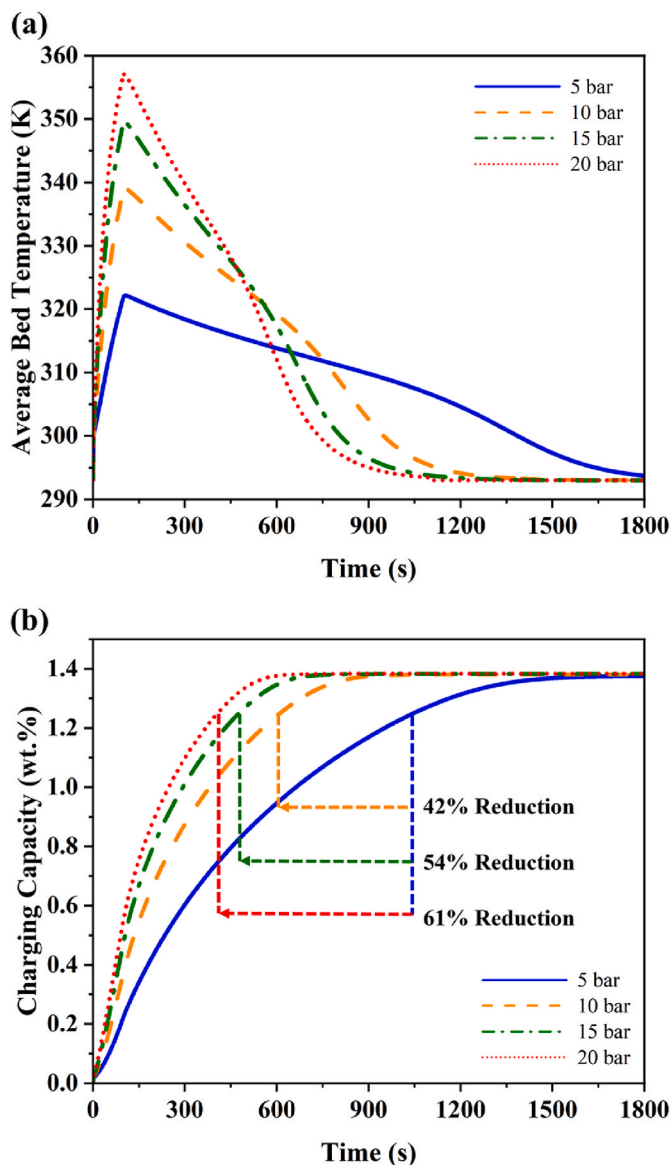


Fig. 10. Temperature and hydrogen charging capacity profiles for different hydrogen inlet pressures: (a) Average bed temperature, (b) Charging capacity profile.

compared to previous parametric studies. The greater the difference between the hydrogen gas pressure and the equilibrium pressure, the higher the reaction rate of absorption, especially at the initial stages. As the hydrogen inlet pressure increases, the difference between the gas pressure and the equilibrium pressure increases, which enhances the absorption reactions. Consequently, the heat generated from the absorption process increases, as shown in Fig. 10 (a). Higher MH bed peak temperatures are observed with higher hydrogen inlet pressures.

A higher initial increase in hydrogen pressure leads to a sharper increase in charging capacity, as shown in Fig. 10 (b). The absorption times to reach 90 % charging are 1033 s, 602 s, 470 s, and 401 s for hydrogen inlet pressures of 5 bar, 10 bar, 15 bar, and 20 bar, respectively. Although the hydrogen inlet pressure was increased in 5 bar increments, the most pronounced reduction in charging time occurred when the pressure increased from 5 bar to 10 bar, resulting in a 42 % reduction. This is because the pressure difference between hydrogen gas and the equilibrium pressure primarily drives the absorption process. However, at higher pressures (above 10 bar), the MH bed temperature rises due to the exothermic absorption reaction, which disrupts the

driving force of absorption. As a result, while increasing the hydrogen inlet pressure reduces the hydrogen charging time, the rate of reduction does not increase linearly. This suggests that increasing the hydrogen inlet pressure up to 10 bar significantly enhances charging capacity but further increases yield diminishing returns.

4.5. Application of copper fin

In this section, the impact of using copper fins instead of stainless-steel fins is analysed, while the reactor itself remains made of stainless-steel. Copper has a thermal conductivity that is approximately 20 times higher than stainless-steel, making widely used material for enhancing heat transfer. As a result, the use of copper fins is expected to significantly improve heat transfer efficiency in the reactor. For a comprehensive comparison between stainless-steel and copper fins, hydrogen charging times for different reactor designs and water flow Re evaluated. The overall comparison between stainless-steel and copper fins is presented in Fig. 11.

The overall hydrogen charging time is notably reduced across all reactor designs and water flow rates when copper fins are used. In the previous results from Section 4.2 Reactor Design, it was showed that stainless-steel fins reduced hydrogen charging time by a maximum of 14 % when comparing Design 1 to Design 4. However, in the same case, copper fins reduced the charging time by 38 %. The effectiveness of copper fins is especially apparent when the surface area of fins is increased. The average reduction in charging time using copper fins is 10 %, 19 %, and 25 % in Design 2,3, and 4, respectively, compared to stainless-steel fins. Notably, when comparing Design 3 and 4, stainless-steel fins showed a reduction of around 3 %, whereas copper fins resulted in a 10 % reduction. These results demonstrate that the use of high thermal conductivity materials, such as copper, inside the MH bed, along with even a slight increase in surface area, is highly effective in reducing hydrogen charging time.

4.6. Optimisation of reactor design

The presence of porous medium (i.e. MH), internal fins, and their arrangements affects the pressure drop in MH bed. This is because the fins act as obstacles, reducing the available hydrogen flow area. Additionally, the increased surface area in the MH bed increases frictional drag. A high pressure drop requires more power to compress the MH bed, and the resulting lower pressure and hydrogen velocity in MH bed can reduce the hydrogen absorption rate. Therefore, selecting the optimal reactor design must carefully consider this trade-off. Fig. 12 (a) presents the hydrogen charging times and corresponding pressure drop for each reactor design. For the comparison, the copper fin with water flow rate in Re 22000 is applied. The pressure drop is measured from the interface of the expansion zone and the MH bed to the bottom of the MH bed, with an inlet hydrogen pressure of 10 bar for all designs. The pressure drop for reactor designs 1 to 4 are 350, 359, 368, and 372 Pa, respectively. As the number of fins increases, a higher pressure drop is observed. In the case of staggered fins (Design 4), hydrogen movement within the MH bed becomes more restricted, leading to a further increase in pressure drop. The corresponding pressure drop is increased to 3 %, 5 %, and 6 % for Design 2, 3, and 4, respectively. On the other hand, the hydrogen charging time is significantly reduced in these designs, decreasing to 83 %, 68 %, and 62 % of the baseline (Design 1). Although pressure drop occur in all designs, the magnitude is approximately 0.04 % of the inlet pressure, making its impact on compression power or reaction rate negligible.

The application of additional fins leads to an increase in the overall reactor mass. The mass increases observed in Design 2, 3 and 4 are 1.6 %, 3.1 %, and 4 %, respectively, compared to Design 1. All reactor designs can store approximately 12 g of hydrogen, releasing about 183 kJ of heat during the absorption reaction. When normalised by the total reactor mass, the heat release rate is expressed as the power-to-mass

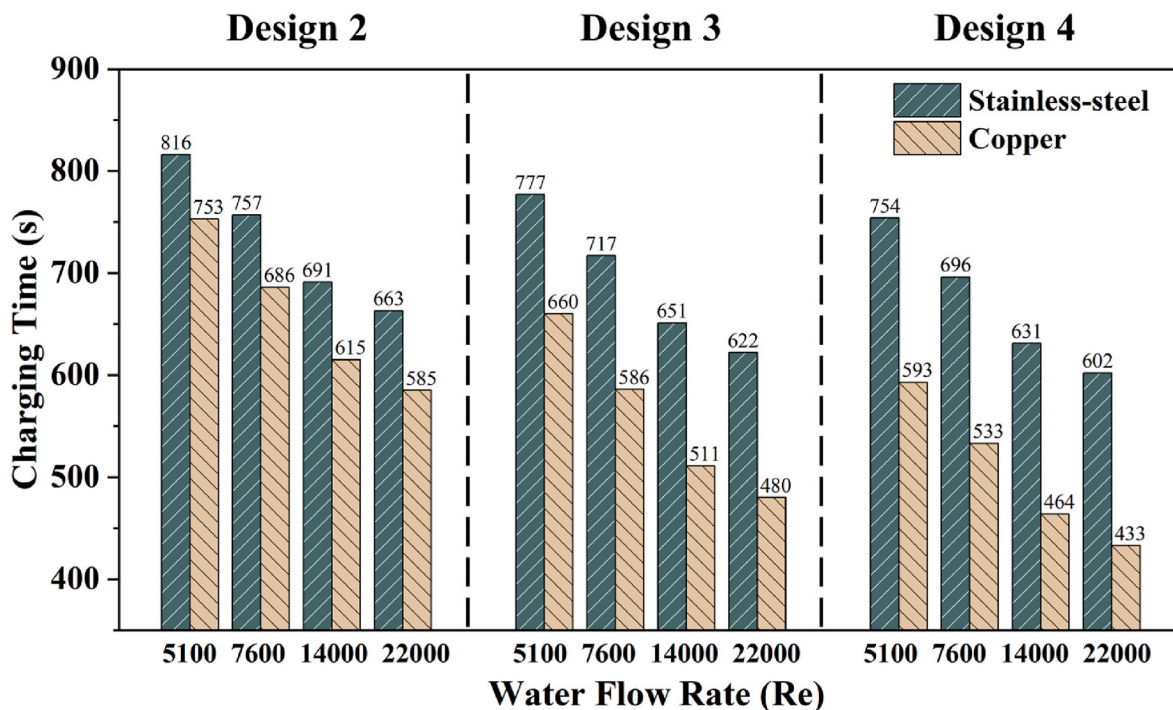


Fig. 11. Hydrogen charging (90 %) time comparison between stainless-steel and copper fins.

ratio (W/kg), calculated by dividing the released heat by the absorption time, as shown in Fig. 12 (b). The power-to-mass ratios for Design 1 to 4 are 153.3, 180.3, 216.4, and 237.9 W/kg, respectively. Notably, while the reactor mass increases with additional fins, the substantial reduction in absorption time allows Design 4 to achieve the highest power-to-mass ratio, a 55 % improvement compared to Design 1. This elevated ratio is particularly beneficial for thermal energy storage and heat pump applications, enabling effective utilisation of the heat generated during the absorption reaction.

From a materials perspective, since MH reactors require relatively high pressure, durable stainless-steel is used for the reactor body. However, replacing the heat transfer fins with copper can effectively reduce absorption time. As a result, applying copper fins in reactor Design 4 enables highly efficient heat transfer and significantly reduces hydrogen charging time.

5. Conclusion

This study conducted numerical simulations of transient heat and mass transfer during MH absorption, focusing on overcoming the challenges posed by the low thermal conductivity of LaNi₅ alloy through reactor design enhancement. The cylindrical reactors with internal cooling fins and an outer water jacket were tested under various boundary conditions to evaluate their performance. Four reactor configurations were studied: 0, 5, and 10 in-line fins and 10 staggered arrangements. The main findings of this study are as follows:

- The application of 10 internal cooling fins in the MH reactor reduced the time to charge 90 % of maximum capacity by 14.6 %. Moreover, using a staggered arrangement of fins further reduced the charging time by 3.2 % compared to the inline arrangement. MH bed temperature and hydrogen charging profiles indicated that cooling fins significantly improved heat dissipation during absorption, enhancing the reaction rate.
- Increasing the Re of water from 5100 to 22000 reduced the hydrogen charging time by up to 21 %. A sensitivity analysis with Re from 5100, to 22000 demonstrated turbulent flow. The SST $k-\omega$

turbulence model was applied for the water flow to accurately capture the turbulent flow characteristics and its effect on MH absorption. The appearance of recirculation flow by the 90° elbow effectively reduced localised temperature, enhancing the reaction rate near this region. HTC improvement was most pronounced at $Re = 14000$, where cooling efficiency significantly increased, while further increase in Re showed diminishing returns in heat transfer enhancement.

- Increasing the hydrogen inlet pressure from 5 bar to 20 bar reduced the absorption time by up to 61 %, with the most significant reduction (42 %) observed when the pressure increased from 5 bar to 10 bar. Loading different hydrogen inlet pressures resulted in significant temperature changes during the MH absorption reaction and a considerable reduction in charging time.
- Replacing stainless-steel fins with copper fins improved heat transfer efficiency due to its higher thermal conductivity. Copper fins reduced charging times by up to 38 %, compared to a maximum reduction of 14 % with stainless-steel fins.
- Despite the increased reactor mass with additional fins, the reduction in absorption time resulted in a higher power-to-mass ratio for reactors with copper fins. This feature makes these designs particularly suitable for applications such as thermal energy storage and heat pump.

This study provides insights into how reactor design modifications, specifically cooling enhancements, can reduce the absorption time of MH. Based on these findings, the proposed reactor design (Design 4 incorporating copper fins) offers the best performance. Future work will focus on experimental studies of metal hydride absorption and desorption using novel heat exchanger designs especially unique fin geometries with a combination of different MH under a broad range of operation parameters. By integrating computational and experimental approaches, our research will provide a more robust understanding of MH hydrogen storage systems, bridging the gap between theoretical and practical implementation. Finally, utilising different machine learning algorithms such as Physics Informed Neural Network (PINN) [42,43] towards optimisation of the reactor/fin design could be utilised.

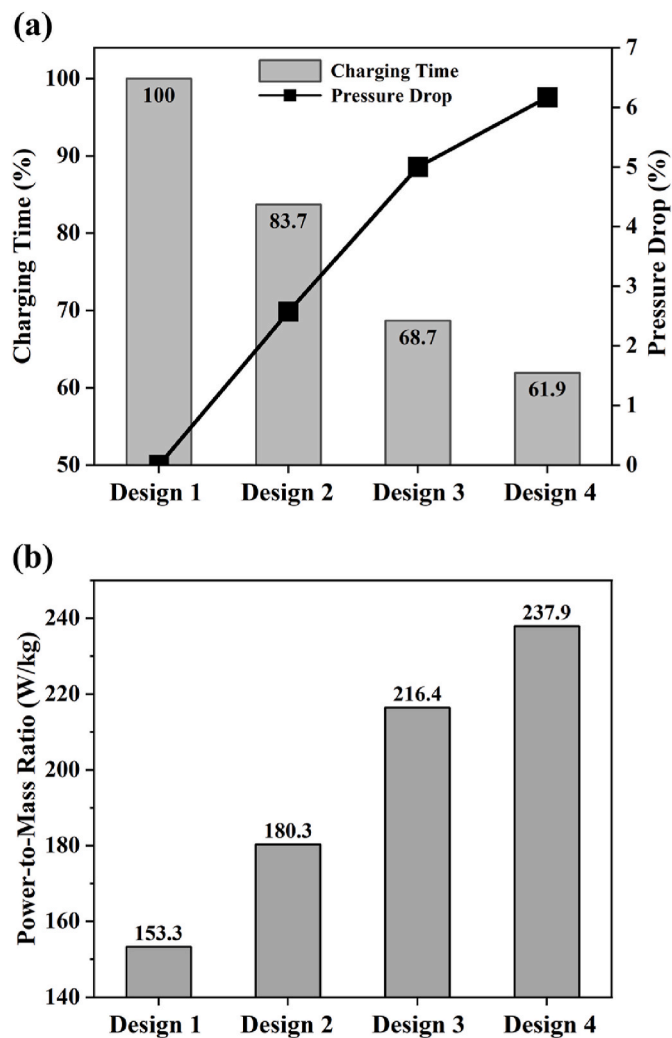


Fig. 12. Evaluation of hydrogen storage performance by reactor designs (a) Charging time and pressure drop and (b) Power generation per reactor mass by absorption.

CRediT authorship contribution statement

Dae Yeob Lee: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Yasser Mahmoudi:** Writing – review & editing, Supervision, Conceptualization. **Vincenzo Spallina:** Writing – review & editing, Supervision, Conceptualization. **Amir Keshmiri:** Writing – review & editing, Supervision, Conceptualization.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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